X03 CCA AND LEAD MANAGEMENT | P

Intent : Mitigate risks of human exposure to chromate copper arsenate (CCA) and lead.

Summary : This WELL feature requires addressing risks associated with human exposure to chromate copper arsenate (CCA) in existing wood structures and lead in soil, playground equipment and artificial turf.

Issue : In the 2000s, the application of CCA (or 'pressure-treated wood') was restricted due to health concerns connecting its presence to arsenic exposure to humans, animals and food crops, and[1] exposure to arsenic is known to cause increased risk of skin, liver, bladder and lung cancers. [2] However, prior to its restriction, CCA was a widely used wood treatment for decades. As a result, CCA presence in our environment still poses as a threat to our health. Additionally, chromates which are catalogued as carcinogenic to humans[3] can be inhaled if CCA-containing wood is ever burnt. Among other existing external hazards is lead, which may be present in paints used on outdoor structures and playgrounds.[4] Lead may also contaminate soils upon chipping, in fibers of artificial turf[5] and in some loose rubber.[6]

Solutions : Identifying and remediating hazards associated with CCA and lead should reduce risk of exposure to and dispersion of contaminants in the environment through leachates. Although not all routes of exposure and bioavailability of lead are fully understood,[6] ingestion and inhalation of lead-containing particles may occur due to contaminated paint or rubber crumbs and, thus, from a precautionary standpoint, testing for lead is recommended.[7]

PART 1 MANAGE EXTERIOR CCA HAZARDS

For All Spaces:

Option 1: CCA assessment and remediation

For all existing wood structures installed before the enactment of laws banning chromated copper arsenate (CCA) which lie outside the building envelope but within the project boundary where human presence is expected (e.g., wooden decks, fences near walkways, playgrounds and outdoor furniture), the following requirements are met:

a. Identify CCA-containing wood through one of the following:
   1. Inspection of purchase records.
   2. Determination of whether legal bans for CCA apply.
   3. Testing for the presence of arsenic in the wood or the soil bearing the wooden structures.
b. Address CCA-containing woods through one of the following:
   1. Dispose of CCA-containing woods following applicable laws, without incinerating nor wood chipping.
   2. Treatment with penetrating (non-film-forming), oil-based, semi-transparent stains that prevent arsenic leaching on a regular basis as recommended by the manufacturer.

OR

Option 2: CCA assessment not required

One of the following is met:

a. All existing wood structures that lie outside the building envelope but within the project boundary where human presence is expected (e.g., wooden decks, fences near walkways, playgrounds and outdoor furniture) were installed after the enactment of laws banning chromated copper arsenate (CCA).
b. The project does not have wood structures that lie outside the building envelope but within the project boundary.
c. The project does not have spaces outside the building envelope but within the project boundary.

PART 2 MANAGE LEAD HAZARDS

For All Spaces:

Option 1: Lead assessment

The project addresses lead hazards through the following:

a. The top 1.5 cm layer in all existing outdoor bare soil (outside the building envelope, post-construction, not covered by grass, vegetation or other landscaping including mulch covered soil) is tested for lead. Each continuous area of bare soil is sampled at least once. If the lead concentration of any sample surpasses 400 ppm by weight,[8] then the following is performed:
   1. A second set of samples is taken at 15 cm, 30 cm, 45 cm and 60 cm deep.[9]
   2. If these samples are above 400 ppm by weight, soil is replaced with soil from another source to the extent of the deepest sample found above this threshold.
b. Lead in existing artificial turf fibers is assessed as follows:[5]
   1. If lead concentration of synthetic turf fibers is unknown, test a sample of fibers to determine the lead concentration using an EPA, ISO or locally accepted protocol.
   2. If the total lead concentration of synthetic turf fibers is greater than 300 mg/kg, perform dust-wipe testing per EPA, ISO or locally accepted protocol for dust-wipe testing to determine the surface dust-lead loading.
   3. If the wipe-testing results show total lead loadings greater than 430 μg/m^2, replace with turf containing lead concentrations less than 300 mg/kg.
c. If existing loose-fill rubber from recycled tires is present on playgrounds, sporting fields, or other surfaces, the

surface is assessed and remediated per the following:

1. Sample the loose-fill rubber using an EPA, ISO or locally accepted protocol for lead testing and perform lead content analysis.
2. If the loose rubber results show total lead loadings greater than 300 mg/kg of rubber, replace the loose-fill rubber.

d. Paint applied to existing playground equipment, installed and painted before the enactment of banning laws, is assessed for lead and removed, as necessary, per the guidance below:

1. Assess the integrity and age of the paint. If the paint is cracked, peeled or chipped collect a sample for laboratory analysis for lead. Follow guidelines and methods described by the World Health Organization[10] or local equivalents for sampling and laboratory analysis.
2. Remove or encapsulate the paint from the playground equipment if the sample contains lead at a concentration over 90 ppm. Removal duties must be performed by a certified specialist or someone with demonstrable experience where no local regulations apply.

e. Newly installed bare soil, artificial turf fibers and rubber fills are certified by the manufacturer or by a third party to not contain lead above the remediation thresholds stated above.

OR

Option 2: Lead assessment not applicable

The following requirements are met:

a. The project does not have existing post-construction outdoor bare soil (e.g., not covered by grass, vegetation or mulch).
b. The project does not have artificial turf.
c. The project does not have loose-fill rubber from recycled tires.
d. Paint applied to existing playground equipment was installed and painted after the enactment of banning laws, or no playground equipment is present.

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